

Shreenath Jaykumar Gandhi

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EDUCATION

Texas Tech University, Lubbock, Texas - Bachelor of Science in Computer Science | Minor: Mathematics | 05/2026 | GPA: 3.5

- Presidential Merit Scholarship Recipient (\$80K)
- Relevant coursework: Computational thinking with Data Science, Artificial Intelligence, Applied Data Science, Reinforcement Learning, OOP, Algorithms, Database Systems, Data Structures, Operating systems, Computer Architecture

SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL, R, Solidity
ML & Data Science: NumPy, Pandas, Scikit-learn, PyTorch, Keras, Statistics, Matplotlib, Seaborn
Data Engineering: ETL, Apache Airflow, Apache Spark (PySpark), REST APIs, Apache Kafka
Databases: MySQL, MongoDB
Web and Frontend: React, FastAPI, HTML, CSS
Cloud and Tools: AWS (S3, Lambda, EventBridge), Docker, Linux, Git/GitHub, Tableau, Apache Kafka
Certifications: META Database Engineer, HackerRank SQL

EXPERIENCE

Machine Learning Research Assistant | Texas Tech Davis College of Agricultural Sciences & Natural Resources 08/2025 – 05/2026

- Process and analyze **LiDAR point-cloud datasets**, segmenting cattle carcass scans into anatomical regions for modeling.
- Engineer **recombination-based augmentation pipelines**, generating **350K+ synthetic samples** to improve model robustness.
- Developing **PointNet++** based **deep learning** model to predict carcass composition from **3D geometric and volumetric features**.

ML Research Assistant | AI for Healthcare | Wu Research Group at Texas Tech University 08/2024 – 05/2026

- Engineering **custom end-to-end deep learning pipelines** for brain **tumor classification** from **medical imaging data**.
- Design **data augmentation strategies** increasing dataset diversity by **4x** and improving model generalization.
- Achieved **98% test accuracy** and **75%** on fully **independent external validation** set **outperforming published baselines (92-94%)**.

PROJECTS

DualBrain AI: Hybrid LLM Routing System with User-Persona Consistency (Python, Ollama, MongoDB, Bcrypt)

- Developed a **hybrid edge-cloud LLM** routing engine that dynamically assigns requests to local or large models based on complexity.
- Designed a **universal account system** storing user traits and preferences ensuring **response quality consistency across LLMs**.
- Implemented **automated prompt engineering** generating **optimized prompts from stored user traits and request context** without manual input.
- Implemented **secure and encrypted user database and authentication** system using MongoDB and Bcrypt.
- Containerized application using **Docker** for consistent deployment across environments.

Daily EOD stock OHLCV tracker pipeline (Python, S3, AWS Lambda, EventBridge, yFinance, Massive, Pandas)

- Built and deployed a **fully automated serverless pipeline on AWS Lambda + EventBridge + S3**.
- Ingests and updates OHLCV data for **4,000+ stocks** every night with **zero human intervention** post deployment.
- Implemented parallel threading (**50 workers**) across **3M+ records**. Entire pipeline updates **under 10 minutes**.

Predicting inflation using Machine Learning (Numpy, Matplotlib, Seaborn, Pandas, Scikit-learn, FastAPI)

- Processed and **aggregated 35+ years of Federal Reserve data** to analyze macroeconomic relationships between interest rates, GDP, unemployment, and inflation.
- Engineered Synthetic Target Rate, Target Deviation features that **improved correlation stability reducing volatility in model predictions** across crisis periods.
- Developed **dual model Random Forest ensemble with LOOCV**, achieving **R²~0.63** and revealing consistent inflation rate divergence.
- **Deployed trained ensemble as a REST API using FastAPI**, exposing a **/predict endpoint** that serves real time inflation predictions from macroeconomic inputs.

ETH Pool: Collateralized Smart Contract ETH Lending Platform (Solidity, HardHat, Node.js) – ETH Denver 2026 Hackathon (BUIDLathon)

- Developed **smart-contract-based lending pool** enabling users to deposit ETH and borrowers to request loans with **150% over-collateralization**.
- Implemented collateral locking, manager-approved loan issuance, repayment tracking, and fund redistribution logic on-chain.

Real-Time Market Risk Monitor (Python, Apache Kafka, Confluent Cloud, PyArrow, NumPy)

- Built production-grade streaming pipeline with independent producer/consumer processes communicating via Confluent Kafka topics.
- Implemented Geometric Brownian price simulator and rolling 50-tick Value at Risk calculator with portfolio level aggregation and threshold alerting.
- Designed batch-optimized Parquet sink writing to Hive-partitioned data lake (partition by ticker) for efficient downstream querying.

MARL Counter-UAS agent for Edge Computing Drone Swarm Interception (Stablebaselines Gymnasium, PPO, PyVista)

- **Architected a custom 50x50x50 environment and 9 parameter overlapping reward-system** to train an agent to intercept drone swarm attacks.
- Achieved **98.7% scenarios where base breach was <=1** in 200 random hostile spawn iterations.
- Led to **emergent layered defense strategy without any hardcoded algorithms/instructions**.
- Leveraged Apple Silicon M4 capabilities by running **32 parallel environments** for training leading to a **robust training of 1B+ iteration in 15 hours**.

INVOLVEMENT AND LEADERSHIP

Treasurer | Students for Global Connections 05/2025 – 05/2026
Student Manager II & Food Protection Manager | Hospitality Services at Texas Tech University 09/2022 – 08/2025
Founding Director of Learning and Development | Major League at Texas Tech 09/2025 – 05/2026